EL NIÑO ACTION BRIEFING · 30 OCTOBER 2026

Zhanshan, Qingdao

The heat forecast for this winter is zero, but in Qingdao cold kills more than twice as many people as heat. Make sure someone knocks on the door of every older person living alone when the cold comes.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Shandong province (province level, ~100 million people)	About zero forecast (−49); 8,900 already past (Jun–Sep)	6-month forecast, Oct 2026 – Mar 2027	Cold (Nov–Mar); then heat in summer 2027

19/19 recommendations verified. 15 corrected, 0 unverified, 4 contraindicated (the four "Do not do this" cards). Exposure figure: Climate Impact Lab El Niño Heat Mortality Tracker data, version of 26 October 2026, pre-peer-review, methods forthcoming. It is published for the whole of Shandong, not for Zhanshan, and it counts heat only, so it cannot see the cold that is this place's main winter risk. The funding target fell from the panel's ¥2,000,000 to ¥355,000 after verification (−82%): the ¥2,000 per household was another district's grant package, the kit costs about ¥500, and Qingdao's city centre sees only about 6–10 snow days a year.

EXECUTIVE SUMMARY — AT A GLANCE



36.07°N 120.39°E, China · Natural Earth

- **The risk.** In Qingdao's climate zone cold causes 12% of deaths and heat 5%, and the forecast "warm" El Niño winter of 2023 still brought record mid-December cold.
- **For you.** Pay the heating fee by 15 November and check your radiators on the 16th; call 96556 if they stay cold. Ignore "10% off" payment offers.
- **The money.** ¥100,000 / ¥355,000 / ¥710,000 (US$14,900 / 52,900 / 105,800); ice grips and grab rails for people over 80 living alone come first.
- **Do not.** Do not use charcoal, a coal brazier, or an unflued gas heater if the heating is late or fails: in January 2026 a Guilin family of three died from a heater with no outdoor flue.*

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* Suggested formulation: check with a health worker before sharing.

FOR YOU, AS A RESIDENT 5 STEPS

In Zhanshan, El Niño matters less than its reputation: the risk this winter is an ordinary cold snap, a radiator that comes on late, and a closed room with a flame in it. The ranked actions below are written for the subdistrict and its partners. These are the ones you can take yourself.

Pay the heating fee by 15 November, and check your radiators on the 16th

PARTIALLY CONFIRMED FROM ACTION 2

BY WHEN	COST TO YOU	PROTECTS
15–16 Nov 2026	The fee on your bill; the call is free	Everyone at home, above all anyone over 75

Qingdao Energy Group's deadline is 15 November and its hotline is 96556. It warns that "九折" or "九五折" (10% or 5% off) offers to pay the fee for you are scams. Across 364 locations in China, each week less of central heating was linked to about 3.3% more winter deaths.

Get a flu vaccine before December at your community health centre

PARTIALLY CONFIRMED FROM ACTION 3

BY WHEN	COST TO YOU	PROTECTS
30 Nov 2026	About ¥86–96 (US$13–14) a dose; ask the centre for its price	People over 60, who suffer over 80% of China's flu deaths

Shandong's free 2026–27 flu vaccine covers only rural residents aged 65 and over, so urban Zhanshan residents pay. The self-pay price is a national figure from October 2026; Qingdao's own price was not found.

If you, or an older neighbour, live alone, give your name and phone number to your grid worker by 15 December

PARTIALLY CONFIRMED FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
15 Dec 2026	None	Older people living alone during a cold wave

* Suggested formulation: check with a health worker before sharing.

Zhanshan already gives SOS bracelets that call 120, mostly to people over 80 who live alone, and ran a daily 8:30 greeting with a same-day visit in one grid. Being on the list is what gets someone to your door. If you have a bracelet, keep it charged.

Check that your gas water heater vents outdoors, never burn charcoal indoors, and replace an electric blanket older than about six years *

PARTIALLY CONFIRMED · FROM ACTION 4

BY WHEN	COST TO YOU	PROTECTS
Before the first cold snap	A check costs nothing; a home gas or CO alarm about ¥150 (US$22) fitted (estimate)	The whole household

Five workers died in Qingdao in January 2005 from a gas water heater that vented indoors. Fire-service staff cite a safe life of about six years for an electric blanket; buy one with the CCC mark, and do not fold it or leave it on unattended. An alarm is a backstop, not a substitute for the flue. *

If you are 65 or over, sign up at an elder canteen for the meal subsidy, and keep three to five days of food in the house

PARTIALLY CONFIRMED · FROM ACTION 7

BY WHEN	COST TO YOU	PROTECTS
Sign up by 30 Nov; food stored by 15 Dec	Sign-up free; about ¥60–100 (US$9–15) of storable food	Meals when ice or a cold snap keeps you in

Shinan extended the meal subsidy to everyone aged 65 and over in December 2025; the city adds ¥3 a lunch and districts add more. Take your ID card and social security card. The Xinyi Zhanshan canteen at 香港西路43号 opened in October 2024; call Zhanshan Community (0532-68888050) to confirm the nearest one.

What to ask of the people in charge

* Suggested formulation: check with a health worker before sharing.

- Ask the Shinan health commission whether it will fund flu vaccine for urban residents aged 75 and over this winter, as Shandong does for rural ones (action 3).
- Ask the subdistrict whether your building is on its heating door-check list, especially if it has no property management (action 2).
- If someone in your household is over 80, disabled, or on low income, ask the subdistrict about the ¥2,000–3,000 (US$300–450) home-adaptation subsidy for grab rails (action 8). The next enrolment for 琴岛e保 health cover is expected in May–June 2027; this year's closed on 30 June.

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet.

1. Turn the existing bracelets and daily greeting into a cold-wave knock rule: no answer means a named person at the door the same day

PARTIALLY CONFIRMED | FRONTLINE · EMERGENCY MANAGER · PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Zhanshan Subdistrict, through its grid workers and volunteer teams	Register by 15 Dec 2026; visits on every orange cold-wave warning, continuing 2–3 weeks after	Existing grid staff and volunteer time	Older people dying at home alone in and after a cold spell

Build a register of residents over 60 who live alone, with their heating type, a phone number, and a named visitor. Zhanshan Community alone has 7 grids, over 200 grid workers, and 23 volunteer teams. In Qingdao, the risk of a cardiovascular death at −4.8 °C was 2.05 times baseline (95% CI 1.55–2.69, 229,288 deaths, 2009–2017), higher in people over 75, and cold deaths come late: the effect lasts 2–4 weeks in Chinese multi-city studies, so visits must continue after the weather turns.

* Suggested formulation: check with a health worker before sharing.

Evidence that check-in systems reduce deaths is mixed: an Italian study found 1.7 fewer deaths per 1,000 people over 65, while a UK telemonitoring trial found no benefit. The daily greeting pilot dates from 2022 and may no longer run; the bracelets cover the whole subdistrict.

2. Check every older person's heating before 16 November in the old walk-ups with no property management

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE · SKEPTIC | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Zhanshan Subdistrict, with Qingdao Energy Group	15 Nov 2026, then a radiator check from 16 Nov	Existing grid staff time	Older people going a cold week without heat because of an unpaid bill or a fault

Confirm the fee is paid (96556) and the radiators warm once heat is on, and warn about the "10% off" payment scam that Qingdao Energy has flagged. Across 364 locations in China, each week less of central heating was linked to about 3.3% higher winter mortality (Hu et al., *Urban Climate* 2022). No early start or temperature trigger for 2026 has been published; Qingdao has started early before, in 2012.

The heating switch-on itself is linked to a 14% rise in weekly deaths in northern China, through coal pollution (Fan, He, and Zhou, 2020). That argues for clean heat and haze advice in mid-November, not for less heating.

3. Fund flu vaccine for urban residents over 75 before December

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Shinan District Health Commission, delivered by community health centres	30 Nov 2026	About ¥86–96 (US$13–14) a dose at self-pay prices; unfunded for urban residents	Flu deaths in the oldest residents, in the season when they peak

Over 80% of China's influenza excess deaths are in older adults, and uptake among people over 60 has been about 4%. A national model put fully public-funded vaccination at about US$4,832 per QALY gained (Yang et al., *BMC Medicine* 2020). Shandong's 2026–27 free dose covers rural residents aged 65 and over, not urban Shinan.

* Suggested formulation: check with a health worker before sharing.

The cost-effectiveness figure is a national model, not a measured reduction in deaths in Qingdao. Shinan has over 170,000 registered residents aged 60 and over; Zhanshan's share was not found.

4. Check gas water-heater flues in older flats, stop charcoal and coal stopgaps, and fit gas or CO alarms as a backstop

PARTIALLY CONFIRMED | SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Zhanshan Subdistrict, with grid workers on the heating visits	Alongside action 2, by 15 Dec 2026	Checks within staff time; alarms about ¥150 (US$22) each fitted (estimate)	Carbon monoxide deaths in closed rooms in winter

Qingdao's own fatal cases came from directly vented gas water heaters: five workers died at 二轻新村 in January 2005, and 19 hotel staff were poisoned the same week. In Guilin in January 2026, a couple and their six-year-old died from a heater with no outdoor flue. Do the flue check on the same visit as the heating check.*

A study of US state laws requiring CO alarms found no protective effect on deaths, though most symptomatic cases in one ice-storm outbreak were in homes without a working alarm. Treat alarms as a backstop; the flue is the fix.

5. Pre-authorise grid workers to warn residents door to door, older people first, on any gas-leak or storm-surge alert, without waiting for formal grading

PARTIALLY CONFIRMED | EMERGENCY MANAGER | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Shinan Emergency Management Bureau, with Zhanshan Subdistrict	30 Nov 2026	None	Hours lost between detecting a hazard and telling the people next to it

In the 2013 Huangdao pipeline explosion, a pressure drop was detected at 02:12 and the blast came at 10:25; 62 people died and 136 were injured. The State Council investigation found the incident was graded routine and residents were not warned or evacuated in time. China's revised Emergency Response Law (in force 1 November 2024, Article 77) already requires residents' committees to begin first

* Suggested formulation: check with a health worker before sharing.

response at once when a situation is urgent. A written decision now makes that duty usable at 2 am.

No published storm-surge evacuation plan for the Taiping Bay coast was found; ask the bureau directly. Evacuation orders remain a government power under the law.

6. Agree now that emergency cash for the poorest elders is paid within 24 hours of an orange cold-wave warning

PARTIALLY CONFIRMED | ECONOMIST · FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Shinan Civil Affairs Bureau, with Zhanshan Subdistrict	Agreed by 15 Dec 2026	Up to ¥3,000 (US$450) per household per case, from the existing temporary-assistance budget	The poorest older people cutting heat or food during a cold wave

Trigger: an orange cold-wave warning from the Qingdao Meteorological Bureau. Qingdao's temporary-assistance rules already allow emergency payment within 24 hours, with paperwork completed within five working days, and subdistricts can approve up to about ¥3,000. In September 2026, urban 特困 support paid ¥1,673 (US$249) a month to 1,162 people citywide, and temporary assistance averaged ¥2,290 a case.

* Suggested formulation: check with a health worker before sharing.

The rules allow "help first, paperwork after", not automatic payment to everyone on a warning, so the subdistrict should pre-list eligible households. Qingdao's price-linked subsidy for low-income households also exists, paying ¥5 per person per month for each point of rise in the low-income price index; it was last found triggered in 2020.

7. Give the elder canteens three days of shelf-stable stock and a standby delivery roster, and use them as an early warning

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Zhanshan Subdistrict and its elder canteens, with Shinan Civil Affairs Bureau	1 Dec 2026	¥30,000–60,000 (US$4,500–8,900), this briefing's own estimate	Housebound elders going without meals for days when roads ice over

Ask canteen staff to flag any regular who stops coming for two days, and treat the meal-delivery list as the first watch list. National food prices are falling (September 2026: food −1.7% on a year, vegetables −8.1%), so the risk is a few days of access, not supply: the January 2016 cold wave pushed fresh vegetable prices up 7.2% in a month, and 29.9% in February.

No Qingdao winter vegetable reserve could be confirmed; a 2012 national rule asks northern cities to hold 5–7 days of consumption.

8. Fit ice grips and bathroom grab rails for people over 80 who live alone, starting with those outside the city subsidy

PARTIALLY CONFIRMED | PHILANTHROPIST · ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Zhanshan Subdistrict, funded through a community charity fund	Before the first snow, usually December	About ¥500 (US$74) a household (estimate: grips about ¥30, two rails fitted about ¥470)	Falls on ice and in the bathroom

* Suggested formulation: check with a health worker before sharing.

In a Wisconsin trial of fall-prone adults over 65, ice grips cut outdoor falls on snow and ice (rate ratio 0.42); about six people fitted prevented one minor injurious fall (McKiernan 2005). Qingdao's home-adaptation subsidy pays ¥2,000–3,000 per household for older people who are over 80, disabled, or on low income; this action covers people who fall outside it.

The trial was in Wisconsin, with many more ice days than Qingdao's city centre (about 6–10 snow days a year), and it prevented minor injuries, not fractures. Grab rails are outside its evidence. Ranked low for that reason.

9. Rehearse a water-supply failure by 31 January, mapping which high-rises lose pressure first

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN · SKEPTIC | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Shinan Emergency Management Bureau, with Qingdao's water utility	31 Jan 2027, if not already done	Staff time	Improvising tanker points and priority lists during a summer shortage

In the 2016–2020 drought, which began in the 2015–16 El Niño, Qingdao's large and medium reservoirs were nearly dry, and Jihongtan reservoir held under seven days of supply. The Yellow River diversion, designed to run about three months a year, set an 846-day continuous run, and in summer more than 90% of supply has come from outside the city. Desalination reached about 340,000 tonnes a day in 2024.

The new Guanlu reservoir (211 million m³) was due for completion at the end of 2025, which weakens a single-reservoir failure. No drought is under way in 2026, and no earlier Shinan water-outage exercise was found.

10. Write the summer 2027 heat plan by April, and add heat to the citywide disaster insurance

PARTIALLY CONFIRMED | HISTORIAN · EMERGENCY MANAGER · PHILANTHROPIST · ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Qingdao Emergency Management Bureau (insurance), with Zhanshan Subdistrict (cooling rooms)	30 Apr 2027	Existing buildings; the insurance change is not priced	Heat deaths in summer 2027, when the counted deaths happen

* Suggested formulation: check with a health worker before sharing.

Shandong had 8,900 extra heat deaths in June–September 2026, and China's National Climate Center has warned of stronger, wider heatwaves in summer 2027. 2023 was Qingdao's warmest year since 1961, with a record 30 days over 30 °C in the city centre. Name community centres as cooling rooms with opening triggers, and point the morning check-in at heat. Qingdao's 2026–2028 disaster insurance already covers everyone in the city, at ¥2 a head, for freezing, snow, typhoon, and flood, paying ¥150,000 (US$22,300) per death; extreme heat is not on its list.

The insurance terms were read from indexed excerpts because the city site could not be reached from outside China; check the full text before relying on them.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Zhanshan Subdistrict	Grid workers and volunteers	Qingdao Energy Group	Shinan Health Commission	Shinan EMB	Shinan Civil Affairs	Elder canteens	Qingdao water utility	Qingdao EMB
1. Cold-wave knock rule	**A**	**R**		**C**	**I**	**C**	**C**		
2. Heating door-check	**A**	**R**	**R**						
3. Flu vaccine for over-75s	**C**	**I**		**A/R**					
4. Flues and CO alarms	**A**	**R**	**C**		**I**				
5. Door-to-door warning order	**R**	**R**			**A**				**C**
6. 24-hour emergency cash	**R**	**C**				**A**			
7. Canteen stock and roster	**A**	**C**				**C**	**R**		
8. Ice grips and grab rails	**A**	**R**				**C**			
9. Water-failure exercise	**C**	**I**			**A**			**R**	
10. Summer heat plan and insurance	**R**	**R**		**C**	**C**				**A**

What the chart shows: Zhanshan Subdistrict is accountable for five of the ten actions and its grid carries the work in eight, so the grid's capacity in December is the binding constraint. Actions 5, 6, and 10 need a written decision rather than new money.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

¥100,000 (US$14,900)

TARGET

¥355,000 (US$52,900)

CEILING

¥710,000 (US$105,800)

LARGEST UNCERTAINTY

Nobody has published how many people over 80 live alone in Zhanshan

Floor: 200 households × ¥500 for ice grips and grab rails = ¥100,000. Target: 500 households × ¥500 (¥250,000) + 500 gas or CO alarms × ¥150 (¥75,000) + canteen stock and roster (¥30,000) = ¥355,000. Ceiling: 1,000 households (¥500,000) + 1,000 alarms (¥150,000) + canteen stock at the high estimate (¥60,000) = ¥710,000; beyond that the grid cannot fit more before the first snow. The household counts are planning

* Suggested formulation: check with a health worker before sharing.

assumptions for a subdistrict of 38,813 people (2020 census). The panel's target was ¥2,000,000; verification cut it by 82%. The funder also advised holding an equal sum for summer-2027 heat check-ins, which sits outside this envelope.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Zhanshan Subdistrict, through a community charity fund (Qingdao Charity Federation runs over 300 community funds citywide; a Zhanshan fund is unconfirmed)	Action 8: ice grips and grab rails for people over 80 living alone outside the city subsidy	¥250,000 (US$37,200)	Designated grant	About ¥500 a household (grips about ¥30; rails priced from Taiwan installation rates as a proxy); McKiernan 2005, RR 0.42 for falls on ice	Partially confirmed
Zhanshan Subdistrict	Action 4: gas or CO alarms in older flats with gas water heaters	¥75,000 (US$11,200)	Designated grant	About ¥150 fitted (gas alarms ¥31–45 retail; CO alarm price estimated); population benefit of alarms unproven	Partially confirmed
Zhanshan elder canteens	Action 7: three days of shelf-stable stock and a delivery roster	¥30,000 (US$4,500)	Pre-positioned stock	This briefing's own estimate; no published cost	Partially confirmed

A foreign organisation can fund this only through a temporary activity filed with public security by its Chinese partner, or through a registered representative office (Overseas NGO Law, Article 9); individuals are not covered by that law. The funder's one thing: the counted deaths here are in summer, and winter money buys fewer fractures and frightening nights that no counter will credit. Spend on winter now, and keep the same grid funded for June.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

Using charcoal, a coal brazier, or an unflued gas heater when the heating is late or fails *

CONTRAINDICATED

A flame in a closed room produces carbon monoxide. Five workers died in Qingdao in 2005 from a gas water heater that vented indoors, and a Guilin family of three died in January 2026 from a heater with no outdoor flue. In Hebei in 2017, coal was removed before the gas supply could meet demand, leaving thousands without heat until coal was allowed again.

Instead: use an oil-filled electric heater, report the fault to 96556, and ask the subdistrict to open a warm room. Never remove one heat source before its replacement works.

Standing down cold-weather preparation because "El Niño winters are mild"

CONTRAINDICATED

In November 2023, during an El Niño, China's National Climate Center forecast a near-normal to warm winter. Mid-December brought North China its lowest temperatures for that time of year since 1961, and Qingdao put 243 snow-clearing teams on standby. The 2026 forecast is "overall above normal" with more cold waves and big temperature swings. A warm average still has cold weeks, and most cold deaths come on ordinary chilly days.

* Suggested formulation: check with a health worker before sharing.

Instead: plan for at least one hard cold wave, with the knock rule (action 1) and pipe-repair crews ready.

Leaving an old or folded electric blanket on all night *

CONTRAINDICATED

Fire-service staff in China cite a safe life of about six years for an electric blanket, and warn against folding them or leaving them on unattended. The risk is highest in older flats with older wiring, where many residents over 75 live.

Instead: warm the bed, then switch off and unplug. Replace a blanket older than about six years, or one without the CCC mark.

Treating household water-saving appeals as the drought plan

CONTRAINDICATED

In the 2016–2020 drought, what kept Qingdao's taps running was outside water: the Yellow River diversion's 846-day run, Yangtze water from 2015, and desalination. Household appeals did not close a gap of that size.

Instead: track reservoir storage and the diversion schedule, and rehearse the supply failure (action 9).

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Jan 2005	Carbon monoxide from a directly vented gas water heater, 二轻新村, Qingdao	5 dead; 19 hotel staff poisoned elsewhere the same week	That heater type had been banned from sale since 2000
Jul 2007	First large green tide reaches Qingdao	None recorded	Annual clean-up; the 2021 bloom was 2.3 times the previous record

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Nov 2013	Huangdao oil pipeline explosion: leak detected at 02:12, blast at 10:25	62 dead, 136 injured	State Council investigation found warning and evacuation were not timely
2015–2020	Drought beginning in the 2015–16 El Niño: rivers stopped, reservoirs nearly dry, Jihongtan under seven days of supply	None recorded	Yangtze water from 2015; desalination about 340,000 t/day by 2024; Guanlu reservoir due end-2025
Jan–Feb 2016	Cold wave during the 2015–16 El Niño	Not counted	National fresh vegetable prices up 7.2% in January and 29.9% in February
Aug 2019	Typhoon Lekima's second landfall, Huangdao coast	5 dead and 7 missing across Shandong; no Qingdao figure found	371,200 people relocated province-wide
2023	Qingdao's warmest year since 1961; a record 30 days over 30 °C in the city centre	Not counted	Nothing found
Dec 2023	El Niño winter forecast mild; mid-December brings North China's lowest temperatures for the period since 1961	None recorded	Qingdao listed homes whose supply pipes freeze repeatedly and insulated them
Jun–Sep 2026	El Niño summer heat	8,900 extra heat deaths in Shandong (Climate Impact Lab)	—

The recurrence nobody has noticed: Qingdao's water crises have come in El Niño years, and each one ended with a bigger outside supply (the Yellow River in 1989, the Yangtze in 2015, then desalination) rather than more storage at home. The deaths that are counted, though, come in summer, and the next El Niño-decaying summer is 2027.

* Suggested formulation: check with a health worker before sharing.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Spend on winter now** (Public health, Frontline, Skeptic) versus **hold money for summer 2027** (Philanthropist, Historian). Cold causes about 12% of deaths in this climate zone against 5% for heat (Chen et al., *BMJ* 2018), but the Climate Impact Lab counts only heat, and its counted deaths fall in summer. This briefing ranks winter actions first because they come first, and keeps the same grid for summer (action 10).
- **Add a cold-wave rider to the disaster insurance** (Economist) versus **the policy already covers freezing and snow** (verification). Heat is the hazard missing from the list, so action 10 proposes a heat rider instead.
- **Daily check-ins save lives** versus **the evidence is mixed**. An Italian study found fewer deaths among people over 65; a UK telemonitoring trial found none. The knock rule is ranked first because the system already exists and costs nothing to point at cold, not because its effect is proven.

WHAT NOBODY COVERED — BLIND SPOT

No lens found a count of people over 80 who live alone in Zhanshan, a storm-surge plan for the Taiping Bay coast, or any rest station for delivery riders and sanitation workers who work outdoors through the winter. Every household figure in the funding section is therefore a planning assumption. The first register under action 1 would replace them, and might show that outdoor workers, not only older residents, belong on the cold-wave list.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

The Climate Impact Lab figure here is published for Shandong province, about 100 million people, not for Zhanshan; it is not a neighbourhood figure. The El Niño projection covers *heat only*; cold, carbon monoxide, water shortage, storms, and food prices are additional to it and are carried here by the historical record rather than by a published number.

* Suggested formulation: check with a health worker before sharing.

- **Scope.** The location was taken as given: the Zhanshan residential district, read as Zhanshan Subdistrict (湛山街道), Shinan District, Qingdao: four communities (湛山, 盐城路, 秀湛路, and 仰口路), 38,813 residents at the 2020 census. Climate Impact Lab figures were available only for Shandong province; no impact region was resolved. Sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Shandong, 8,900 additional heat deaths June–September 2026, −49 forecast October 2026 – March 2027, 8,900 in total; China, 85,200. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only; we inferred this from the data, because cold countries show exactly zero in every winter month.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources in Chinese and English, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Five verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted. Its "Download Inventory" link leads to a registration form, and the browse page loads its data in the browser. No action here should be read as having no precedent; none was checked.

* Suggested formulation: check with a health worker before sharing.

- **Money.** Costs are shown in Chinese yuan with US dollars in parentheses, at ¥6.7138 = US$1 (People's Bank of China central parity, 30 October 2026).
- **Reader.** This briefing was built at El Niño Ready's initiative, not from a request. No role was given, so the For you section is written for a resident. The "For you" steps were taken from the verified actions and from what the panel found about residents here; any step not drawn from a ranked action was verified on the same three tests. The rest of the report is written for the subdistrict and its partners.
- **Health wording.** Passages where a wording error could hurt someone (heaters and carbon monoxide, gas water heaters, electric blankets, and fire) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Several Qingdao government and newspaper sites could not be reached from outside China, so some figures (the disaster-insurance terms, the canteen subsidy amounts) rest on indexed excerpts. Grab-rail prices use Taiwan installation rates as a proxy. The locator map shows Qingdao at country scale; the subdistrict itself is not drawn.

References · status after verification

Climate Impact Lab, El Niño Heat Mortality Tracker data v261026 (downloaded 30 Oct 2026); tracker · confirmed

Population: citypopulation.de (2010, 2020 census); 湛山街道; Zhanshan Community: 12371.cn, 2023; elder services and pilots: People's Daily, 2022 · corrected

Qingdao cardiovascular cold study (229,288 deaths): Int J Environ Health Res · confirmed (abstract); Chen et al., BMJ 2018: PMC6207921 · confirmed; heating and mortality, Hu et al. 2022: Urban Climate · confirmed (abstract); heating switch-on: Fan, He, and Zhou 2020 · confirmed (working paper)

Cold lag: four Chinese cities; Tianjin · confirmed

Heating fee and scam warning: Qingdao Energy Group notice, 13 Oct 2026 · confirmed; season dates: ifeng · confirmed (local media)

Flu vaccine: Yang et al., BMC Medicine 2020; Shandong 2026–27 free scheme; self-pay prices · corrected

Carbon monoxide: Qingdao 2005 deaths; Guilin 2026 investigation; US alarm-law study; gas alarm prices: ZOL · corrected

Huangdao 2013: State Council investigation report · confirmed; Emergency Response Law 2024: full text · confirmed

Temporary assistance and safety-net figures: ifeng, Sept 2026; Qingdao temporary-assistance notice, 2020; price-linked subsidy: 青价综〔2016〕38号 · corrected

* Suggested formulation: check with a health worker before sharing.

Disaster insurance 2026–2028: Qingdao government; Shandong plan · corrected (read from excerpts)
琴岛e保 enrolment: Qingdao Evening News · confirmed (closed 30 June); home-adaptation subsidy: Qingdao Civil Affairs, 2024 · corrected
Overseas NGO Law: ChinaFile translation; community funds: Qingdao briefing, Sept 2024 · corrected
Ice grips: Cochrane table (McKiernan 2005); Qingdao snow days: Qingdao Evening News · corrected
Elder canteens and meal subsidy: Qingdao Daily, 19 Dec 2025; CCTV, 2024 · corrected
Food prices: NBS CPI, September 2026; January–February 2016 vegetables; 2012 reserve rule · confirmed
Water: Jihongtan under seven days; 846-day diversion; desalination 340,000 t/day; Guanlu reservoir · corrected
Do-not sources: AFP, Hebei 2017; Caixin; electric blankets; December 2023 cold; CMA winter review · corrected
Outlook and record: National Climate Center, 29 Sep 2026; 2023 warmest year; green tide 2021; Lekima · confirmed
Generated 30 October 2026 for Zhanshan, Qingdao. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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